KALAN FERGUSON

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Australian Citizen | Eligible to obtain Defence Security Clearance (NV1) upon nomination

CAREER OBJECTIVE

Penultimate-year Mechanical Engineering (Honours) student at QUT with an Aerospace minor (Major GPA 6.0/7.0, Executive Dean's Commendation). My work spans first-principles structural integrity, FEA fatigue validation, drivetrain kinematics, and thermo-fluid systems: a multidisciplinary foundation applied through physical hardware testing and root-cause failure investigation. Seeking a 2026/2027 engineering internship where rigorous analysis and data-driven component validation deliver reliability outcomes across defence, energy, and advanced manufacturing platforms.

EDUCATION

Bachelor of Mechanical Engineering (Honours) | Aerospace Minor | QUT *Expected Dec 2027*

- Major GPA: 6.0/7.0 | Cumulative GPA: 5.7/7.0
- QUT Executive Dean's Commendation for Academic Excellence (2025)
- High Distinctions: Thermodynamics, Foundations of Electrical Engineering, Fundamentals of Mechanical Design, Engineering Mathematics & Statistics
- Current units: Fluid Dynamics | Mechanical Systems Design | Research in Engineering Practice | Unmanned Aircraft Systems

TECHNICAL PROJECTS

3-Stage Planetary Gearbox | Design, Analysis & Physical Testing *2025*

- Designed, built and tested a 3-stage planetary gearbox to a 20 Nm operational torque target, running it to controlled failure to validate analysis against real hardware behaviour
- Conducted post-test failure analysis, tracing sequential gear-tooth fatigue to localised housing compliance deformation rather than material strength
- Applied Lewis tooth bending equations and contact stress analysis to assess dynamic fatigue allowances across all stages
- Recommended dual-sided carrier plates, increased face width, and housing stiffening: reliability-engineering logic applicable to torque-transmission systems under sustained heavy load

Weld Fatigue Design Audit | EGB316 Design of Machine Elements *2026*

- Conducted a full fatigue-life audit of a critical welded joint using first-principles analysis and ANSYS FEA, demonstrating infinite fatigue life (FoS = 6.33)
- Characterised a 100×50 mm fillet weld group; applied Modified Goodman analysis (Kf = 2.7, Marin-modified endurance limit) for a yield FoS of 14.46
- Ran mesh convergence from 1 mm to 0.1 mm at the weld toe; realistic boundary conditions reduced peak stress by 40% versus a fixed constraint, quantifying the conservatism in the hand solution
- Validated against AS 4024 (Weld Quality Grades) and AS/NZS structural steel standards

Fixed-Wing Autonomous UAV | QUT Aerospace Society, Aerostructures Co-Lead *2025 – Present*

- Co-led structural design and configuration trade studies for a blank-sheet autonomous UAV, from concept through detailed design (CAD)
- Sized wing planform and internal load path (carbon-tube main spar, D-box torsional stiffening) against competing constraints: hand-launch stall speed and a fixed span limit
- Evaluated boom-and-pod, flying wing, and conventional layouts against stability, structural efficiency, and manufacturability; recommended NACA 4412 at Re ≈ 142k

F-35B 3-Bearing Swivel Duct | Independent Kinematic Study *2025*

- Independently designed and motion-validated a fully parametric SolidWorks model of the swivel duct's rotational synchronisation across its thrust-deflection range
- Resolved mechanical interference and geometric clearance constraints through iterative refinement of the bearing interfaces

PROFESSIONAL EXPERIENCE

Installer Assistant | Brisbane Glass Splashbacks *Sep 2024 – Aug 2025*

- Held ±1 mm tolerances across 300+ custom installations; resolved on-site fit-up and alignment constraints under real conditions, zero rework

Operations & Logistics Coordinator | LSKD Loganholme *Nov 2024 – Jan 2025*

- Sustained 100% KPI compliance across 5,000+ order fulfilment cycles under high-tempo seasonal operations

TECHNICAL SKILLS

CAD & Design: SolidWorks (parametric modelling, assemblies, motion studies), engineering drawings
FEA & Simulation: ANSYS Mechanical (mesh convergence, boundary condition validation), SolidWorks Simulation
Analysis Methods: Modified Goodman fatigue, Von Mises, Lewis tooth bending, kinematic analysis, load-path and failure-mode analysis
Programming: Python, MATLAB
Standards: AS 4024 weld quality grades, AS/NZS structural steel